# Bayesian Age-Period-Cohort Modeling

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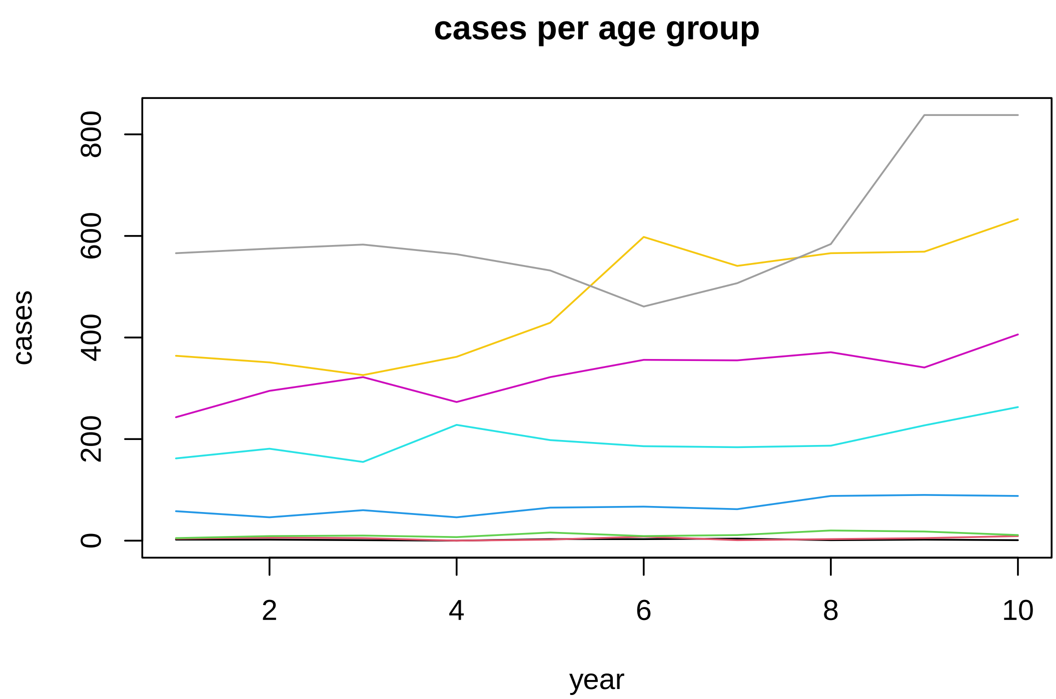
## Data example

BAMP includes a data example.

```
data(apc)
plot(cases[,1],type="l",ylim=range(cases), ylab="cases", xlab="year", main="cases per age group")
for (i in 2:8)lines(cases[,i], col=i)
```




## APC model with random walk first order prior

```
model1 <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
               periods_per_agegroup = 5)
```

bamp() automatically performs a check for MCMC convergence using Gelman and Rubin's convergence diagnostic. We can manually check the convergence again:

```r
checkConvergence(model1)
```

```
## [1] TRUE
```

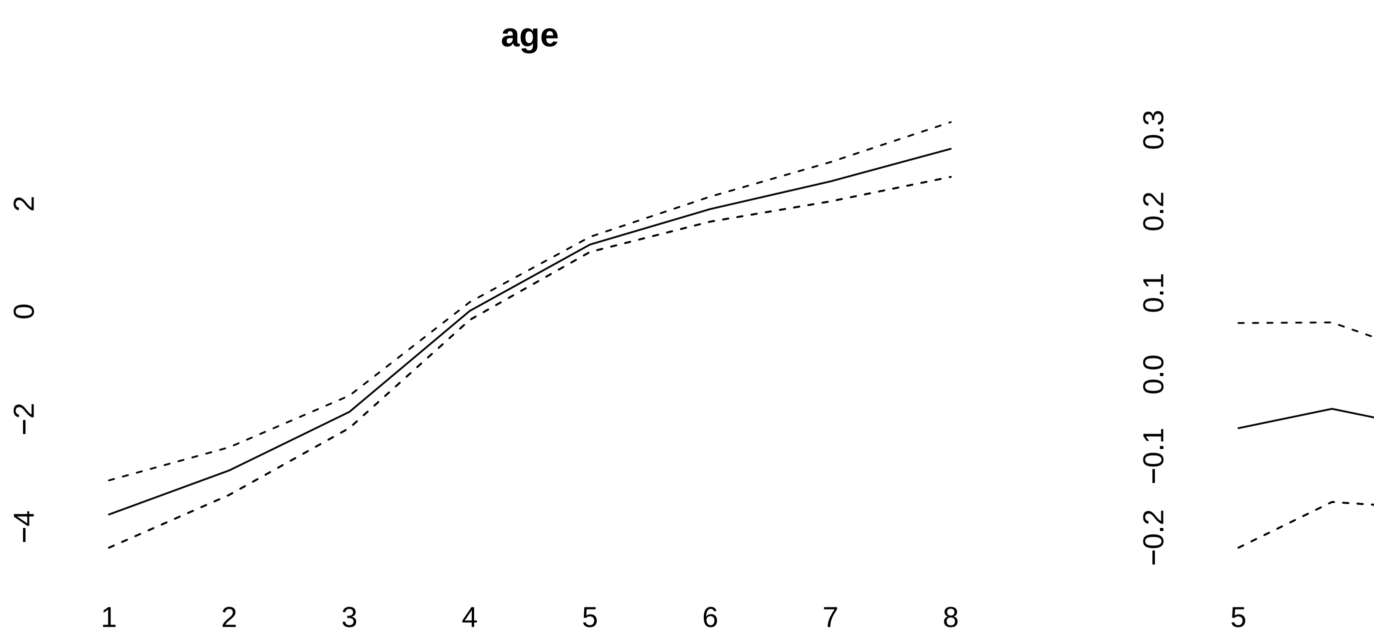
Now we have a look at the model results. This includes estimates of smoothing parameters and deviance and DIC:

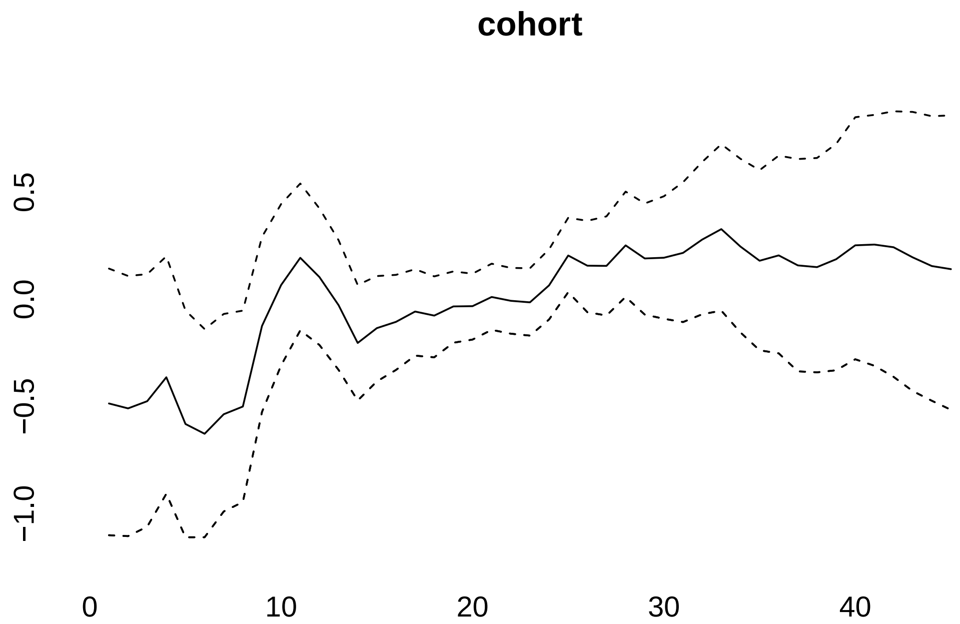
```r
print(model1)
```

```
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     231.22
## pD:            36.93
## DIC:          268.15
##
##
##  Hyper parameters:                    5%          50%          95%
## age                                0.364        0.900        1.907
## period                            68.903      198.279      621.735
## cohort                            33.980       59.834       95.958
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

We can plot the main APC effects using point-wise quantiles:

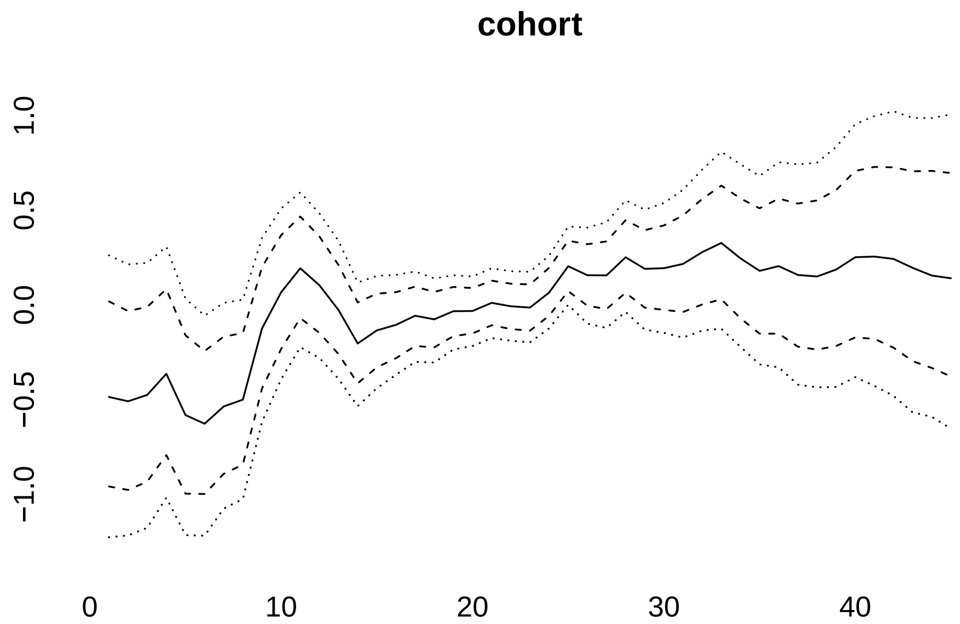
```r
plot(model1)
```






More quantiles are possible:

```
plot(model1, quantiles = c(0.025,0.1,0.5,0.9,0.975))
```




### APC model with random walk second order prior

```
model2 <- bamp(cases, population, age="rw2", period="rw2", cohort="rw2",
              periods_per_agegroup = 5,
              mcmc.options=list("number_of_iterations"=200000, "burn_in"=100000, "step"=50, "tuning"=500
              hyperpar=list("age"=c(1,.5), "period"=c(1,0.05), "cohort"=c(1,0.05)))
```

```
##
## Automatic check procedure removed 1 Markov chain. Please check for convergence using checkConvergence
```

```
checkConvergence(model2)
```

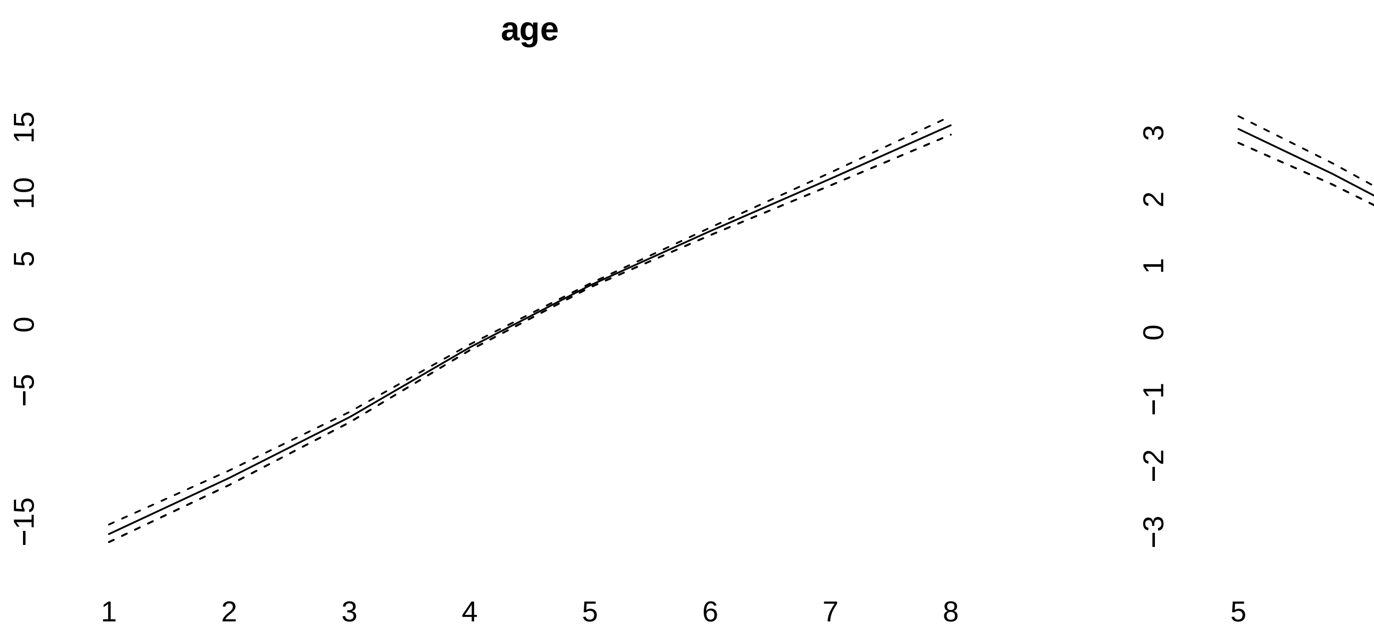
```
## [1] TRUE
```

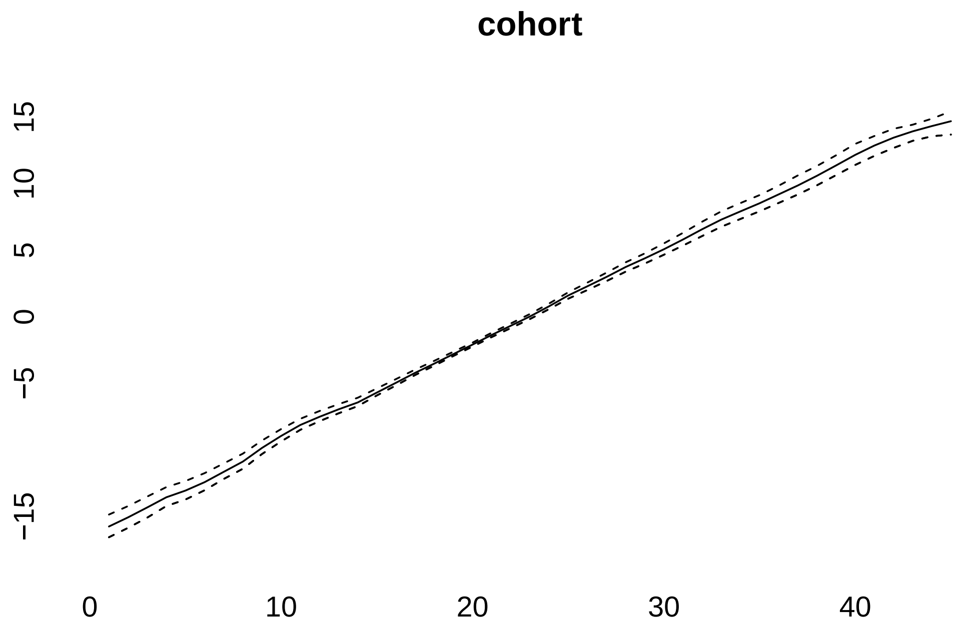
```
print(model2)
```

```
##
##  Model:
## age (rw2)  - period (rw2)  - cohort (rw2) model
## Deviance:     233.98
## pD:            37.14
## DIC:          271.11
##
##
```

```
##  Hyper parameters:                    5%          50%          95%
## age                               0.995        2.908        6.598
## period                           16.211       41.163       91.103
## cohort                           22.444       43.666       79.953
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```r
plot(model2)
```






```r
model3<-bamp(cases, population, age="rw1", period=" ", cohort="rw2",
             periods_per_agegroup = 5)
```

```r
checkConvergence(model3)
```

```
## [1] TRUE
```

```r
print(model3)
```

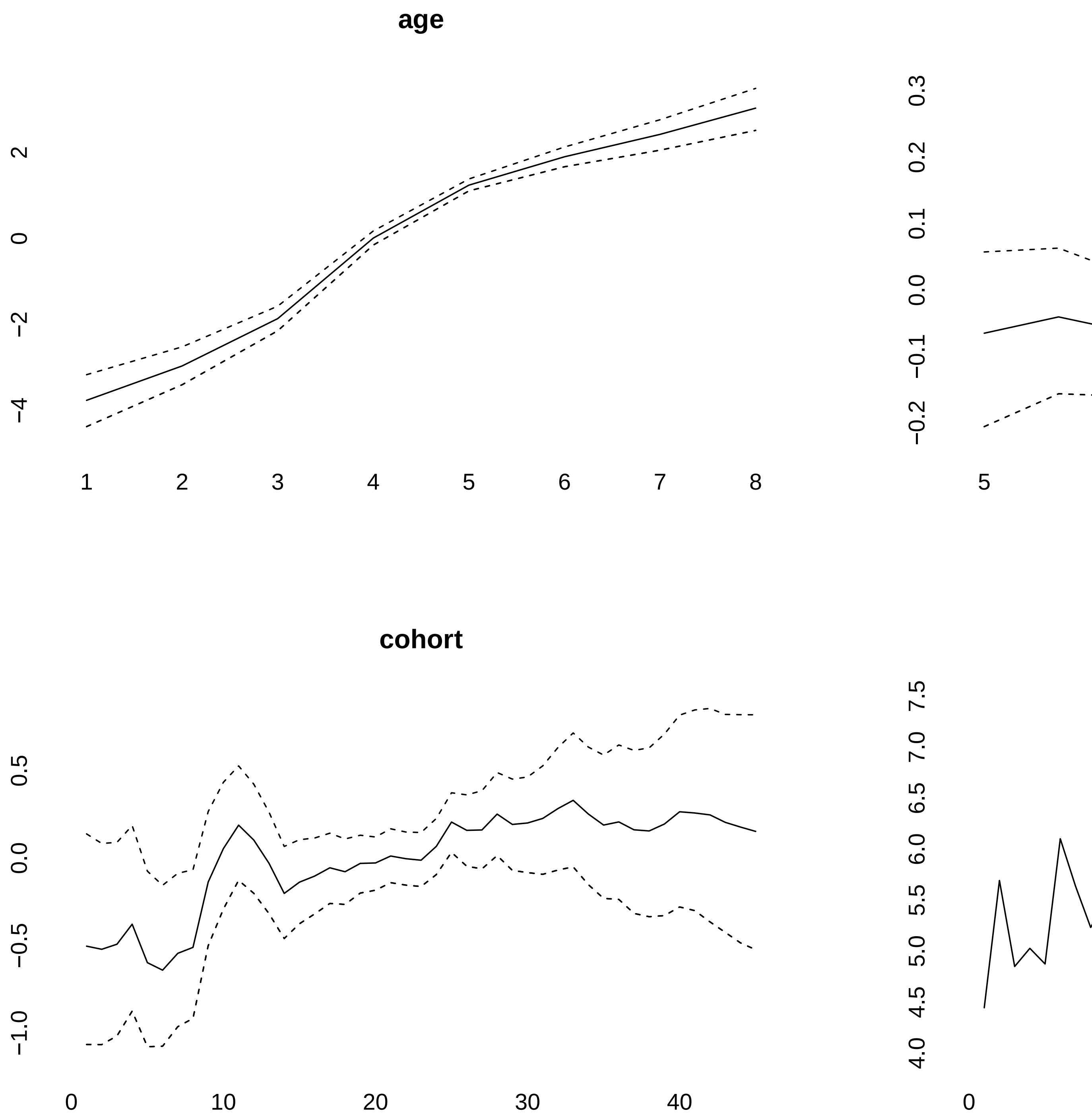
```
##
##  Model:
## age (rw1) cohort (rw2) model
## Deviance:     276.66
## pD:            30.18
## DIC:          306.85
##
##
##  Hyper parameters:                  5%          50%          95%
## age                              0.289        0.727        1.493
## cohort                          38.542       74.001      138.677
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

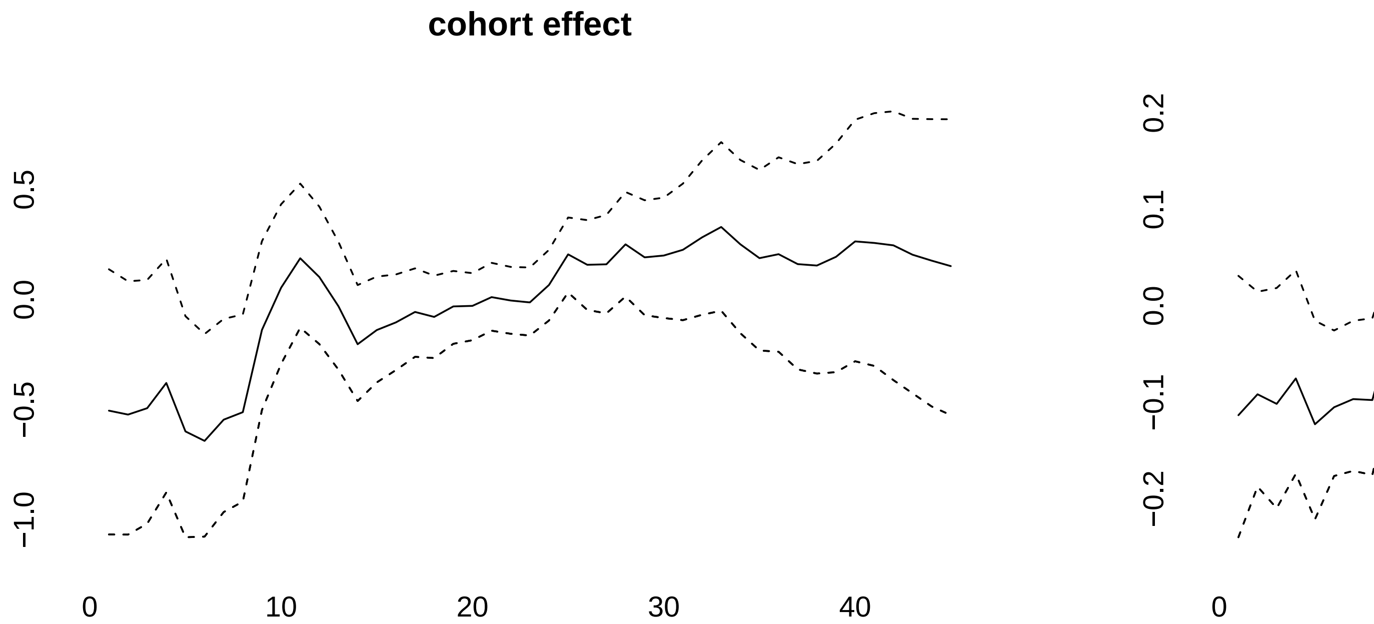
```r
plot(model3)
```


```
(model4<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
             cohort_covariate = cov_c, periods_per_agegroup = 5))
```

```
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     231.33
## pD:            36.80
## DIC:          268.13
##
##
##  Hyper parameters:                   5%          50%          95%
## age                               0.363        0.937        1.989
## period                           67.003      195.378      596.195
## cohort                           35.661       60.503       97.350
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
plot(model4)
```


age
cohort

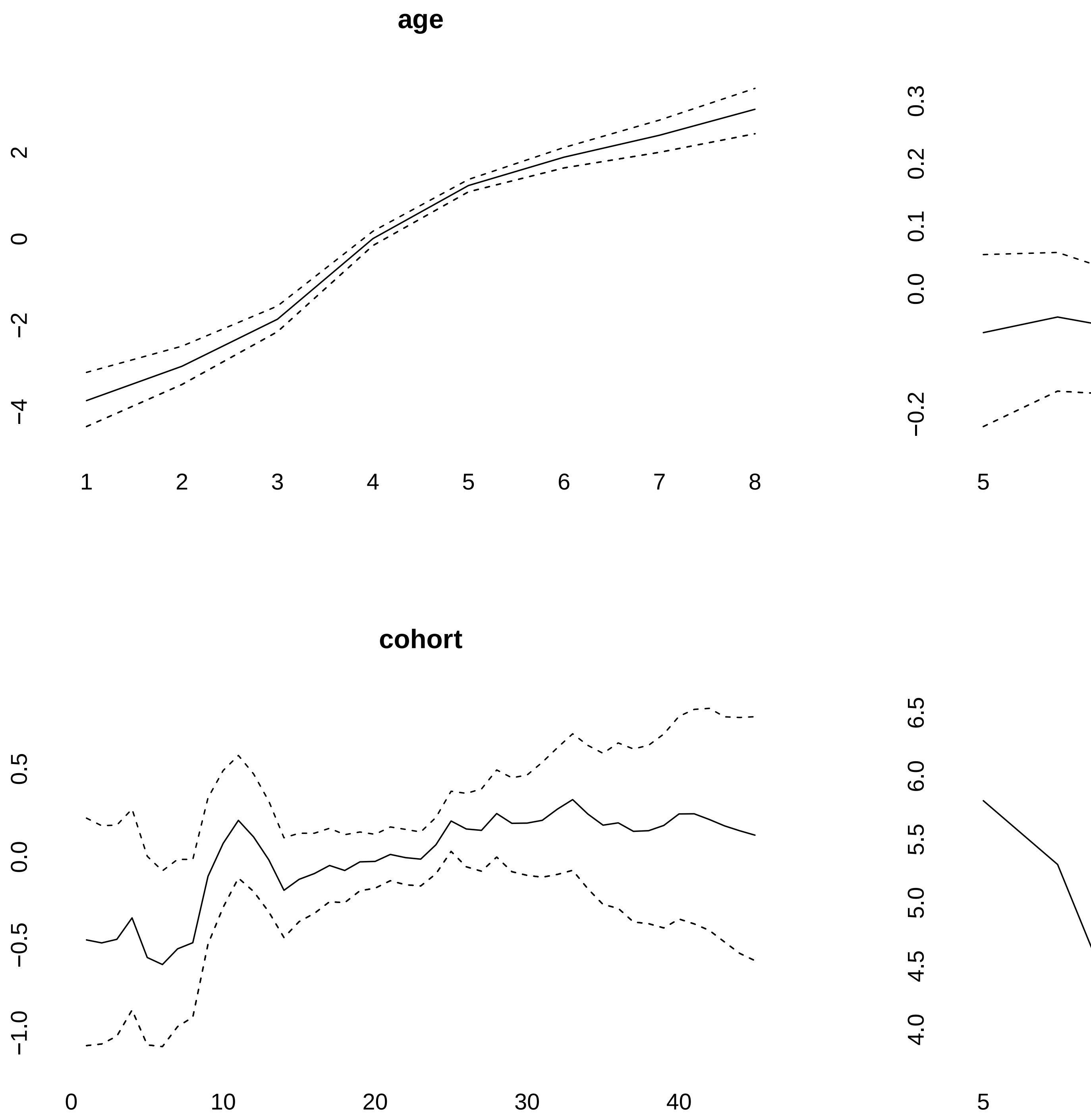


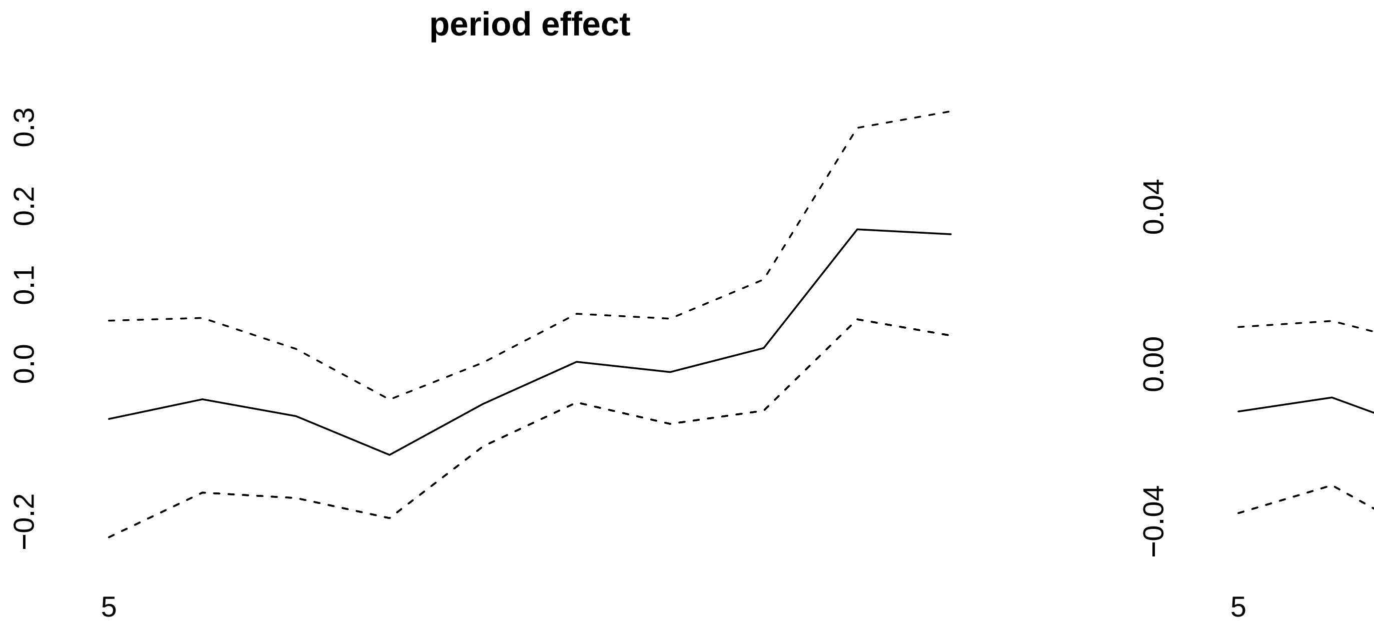

```
(model5<-bamp(cases, population, age="rw1", period="rw1", cohort="rw1",
             period_covariate = cov_p, periods_per_agegroup = 5))
```

```
##
## Automatic check procedure removed 1 Markov chain. Please check for convergence using checkConvergenc
## Warning: MCMC chains did not converge!
```

```
##
## WARNING! Markov Chains have apparently not converged! DO NOT TRUST THIS MODEL!
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     231.42
## pD:            36.91
## DIC:          268.34
##
##
##  Hyper parameters:                    5%          50%          95%
## age                                0.363        0.940        1.994
## period                            65.471      198.174      601.858
## cohort                            34.803       59.715       97.556
```

```
plot(model5)
```


age
2
0
−2
−4
1
2
3
4
5
6
7
8
0.3
0.2
0.1
0.0
−0.2
5
cohort
0.5
0.0
−0.5
−1.0
0
10
20
30
40
6.5
6.0
5.5
5.0
4.5
4.0
5



period effect
0.3
0.2
0.1
0.0
−0.2
5
0.04
0.00
−0.04
5